

MongoDB

MongoDB Date:12-05-2026 Task

Update Queries with Projection Context

1. **Update** the **salary** to **₹95,000** for the employee whose **name** is "Tanvi Thakur". Then display only the **name**, **salary**, and **job_title** of that employee.
2. **Update** the job title to "Senior Software Engineer" for all employees in the **IT department** with more than 5 years of experience. After the update, retrieve their **name**, **department**, and **job_title**.
3. Increase the **salary** by **₹5,000** for employees from the **Sales department** whose current **salary** is less than **₹60,000**. Then list only their **name**, **salary**, and **department**.
4. Change the **department** to "**Engineering**" for employees living in Jaipur or Pune and having the job title "Frontend Developer". **Display** their **name**, city, and new **department**.
5. Set the **bonus** field to true for all employees whose experience is greater than or equal to 8 years and **salary** exceeds **₹100,000**. **Project** their **name**, **experience_years**, **salary**, and **bonus**.

Delete Queries with Pre-Check Projection

1. **Delete** all employees who are in the **Finance department** and have less than 2 years of experience. Before deleting, show their **name**, **department**, and **experience_years**.
2. **Delete** one employee whose **salary** is exactly **₹50,518**. First display their **name**, **salary**, and **job_title**.
3. Remove all employees who live in Amritsar and have the job title "Database Administrator". **Project** their **name**, **address.city**, and **job_title**.
4. **Delete** all employees who have the skill "**MATLAB**" and are from the **R&D department**. Before deletion, show **name**, **department**, **skills**.
5. **Delete** employees whose **salary** is below **₹45,000** or who have less than 1 year of experience. **Display name**, **salary**, and **experience_years** before deletion.

Count Queries with Contextual Field Checks

1. **Count** how many employees are in the **Marketing department**. Also show a sample list of their **name** and **department**.
2. **Count** employees who earn more than **₹90,000** and have more than 6 years of experience. **Display** only **name**, **salary**, and **experience_years** of matching entries.
3. **Find** the number of employees who have the skill "**JavaScript**". **Show** their **name** and **skills**.

4. **Count** the number of employees who are younger than 30 and from either **IT** or **HR** department. **Show** their **name**, **age**, and **department**.
5. **Count** how many employees are not from the **Sales** or **Operations** departments. **Project** only **name** and **department**.

Comparison & Logical Operator Queries with Projection

1. **Find** employees whose **salary** is between ₹70,000 and ₹90,000 and have **"React"** as one of their **skills**. **Show** only **name**, **salary**, and **skills**.
2. **Retrieve** all employees who either have the job title "DevOps Engineer" or live in Nagpur. **Display** **name**, **job_title**, and **address.city**.
3. Get employees who are not from **"Marketing"** and also do not have **"IoT"** as a skill. **Show** their **name**, **department**, and **skills**.
4. **Find** employees whose experience is greater than 5 years, **age** is under 30, and are from the **IT** department. **Project** **name**, **age**, **experience_years**, and **department**.
5. **Find** employees who are either from the **HR** department or from cities other than Pune and Delhi. **Display** **name**, **department**, and **address.city**.

MongoDB Date:13-05-2026 Task

1. Use **\$match** to filter employees in the **'Marketing'** department and **\$group** them by **job_title** to count how many employees are in each role.
2. Match employees whose **salary** is greater than ₹80,000 and **group** them by **department**, calculating the **average experience_years** in each department.
3. Match employees who have the **skill 'React'** and **group** them by **address.city**, showing how many such employees live in each city.
4. Filter employees whose **experience_years** is greater than 5 and **group** them by **department**, calculating the **maximum salary** in each group.
5. Match employees who belong to the **'IT'** or **'Operations'** departments, then **group** by **department** to find the **sum of their salaries**.
6. Match employees whose **job_title** contains the word **'Engineer'** and **group** by **job_title** to get the **total number of employees** in each.
7. Match employees living in **'Pune'** or **'Jaipur'** and **group** them by **address.pincode**, showing the **average age** per pincode.

8. Match employees with more than **3 skills** and **group** by **department**, counting how many such **multi-skilled employees** each department has.
9. Match employees whose **salary** is less than **₹50,000** and **group** them by **job_title** to find the **minimum experience** in each role.
10. Match employees who have '**IoT**' in their **skills** and **group** them by both **department** and **job_title**, counting the **total employees** for each unique combination.
11. **Join** each customer with their **orders** and calculate the total amount spent and total number of **orders**. **Sort** the results in descending order of total amount spent, and return only the top 5 customers.
12. List all customers along with their **minimum**, **maximum**, and average order amounts. Return only those customers who have placed more than 1 order.
13. Generate a **report** showing the number of **orders** and total revenue for each customer who placed **orders** in **2025**. Store the output in a new collection called **customer_2025_summary**.
14. **Join** customers with their **orders**, and project only those who placed **orders** in March **2025**. Show their **name**, **email**, and total amount spent in that **month**. 5. For each customer, return their **name**, **email**, and a new field called **lastUpdated** containing the current **timestamp**. Also, include their total number of **orders** and total amount spent.
15. From the **orders**, **group** data by the **year** and **month** of the order date. Calculate the total revenue, total number of **orders**, and the highest order amount in each **group**. 7. Find the customers who placed **orders** in **2025**, and for each, show their **name**, total amount spent, and number of **orders**. Return only the first 10 records after **skipping** the first 5.
16. **Join** customers with **orders**, then **group orders** by customer and show only those customers who spent a total of more than ₹8000. Include their **name**, total amount, and number of **orders**.
17. **Join** customers with their **orders** and calculate for each customer: the average order amount, total number of **orders**, and highest order value. Then sort the customers by average order amount in descending order.
18. Create a **monthly** customer order summary by **grouping orders** based on customer and the **month** of the order. For each **group**, calculate total amount spent and number of **orders**. Export the result to a collection named **monthly_orders_summary**.